

Model Validation of a Damped Simple Pendulum: Period Scaling, Amplitude Effects, and Symmetry

Julian Woo*

PHY324 University of Toronto

Abstract

In this experiment we tested whether a self devised pendulum follows the standard underdamped-oscillator model: a constant period independent of amplitude and mass, exponential decay of the oscillation with time constant τ , and the small-angle scaling $T \propto \sqrt{L}$. A simple pendulum was constructed with three effective lengths and three release angles, and its motion was recorded using video tracking software to obtain $\theta(t)$. Periods were extracted from successive angular maximums (cycle-by-cycle T_i) and summarized by a mean period for each run, with uncertainties built from the scatter across cycles and the video frame time step; damping was quantified by fitting the peak-amplitude envelope to $A(t) = A_0 e^{-t/\tau}$. Over $\theta \approx 0.15$ to 2.0 rad, the data is consistent with no detectable amplitude dependence, with a weighted constant fit $T_0 = 1.15$ s and $\chi^2_\nu = 1.14$. Exponential envelope fits gave $\tau \sim 15$ s (e.g., $\tau = 15.4 \pm 0.4$ s and $\tau = 14.8 \pm 0.460$ s for two conditions at one length). In contrast, the length-scaling test refuted the $\sqrt{L}$ prediction as implemented here: pairwise period ratios were $R_{30,2,34,5} = 0.954 \pm 0.003$, $R_{30,2,37,1} = 0.927 \pm 0.003$, and $R_{34,5,37,1} = 0.972 \pm 0.003$, indicating a systematic disagreement most likely dominated by effective length and geometry systematics in the apparatus.

1 Introduction

A simple pendulum is a system for testing how well an idealized model of oscillatory motion describes a real mechanical system. In the ideal limit of a point mass suspended from a massless string moving in a single vertical plane, the motion is periodic and can be treated as approximately harmonic when the angular displacement is small. In practice, real pendulum experience dissipation and can depart from purely harmonic behaviour, especially at larger angular amplitudes. The purpose of this project was to measure the pendulum's period and damping time scale and, more importantly, to identify where the standard damped-oscillator description is consistent with the data and the where it fails for a self built pendulum.

The equation of motion follows from torque balance about the pivot. Using $\sum \tau = I\ddot{\theta}$ with moment of inertia $I = mL^2$ for a point mass m at distance L , the gravitational torque is $\tau_g = -mgL \sin \theta$. To model dissipation, a damping torque proportional to angular speed is used, $\tau_d = -\gamma\dot{\theta}$. The resulting equation of motion is

$$mL^2 \ddot{\theta} + \gamma \dot{\theta} + mgL \sin \theta = 0. \quad (1)$$

For sufficiently small angles ($|\theta| \ll 1$ rad), $\sin \theta \approx \theta$ and Eq. (1) reduces to the linear damped-harmonic-oscillator form,

$$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = 0, \quad \omega_0^2 = \frac{g}{L}, \quad \beta = \frac{\gamma}{2mL^2}. \quad (2)$$

In the underdamped system ($\beta < \omega_0$), the solution is an exponentially decaying sinusoid,

$$\theta(t) = \theta_0 e^{-\beta t} \cos(\omega_d t + \phi_0), \quad \omega_d = \sqrt{\omega_0^2 - \beta^2}, \quad (3)$$

so the amplitude envelope decays exponentially while the oscillation period is approximately $T \simeq 2\pi/\omega_d \approx 2\pi\sqrt{L/g}$ when damping is weak. Writing the decay as $e^{-t/\tau}$ shows the decay constant τ with $\tau = 1/\beta$ in this model.

We will test the following predictions implied by the damped, small-angle model: (i) the oscillation envelope is consistent with a single exponential decay characterized by one time constant τ ; (ii) the period T is independent of the initial amplitude θ_0 and of the bob mass; and (iii) the period scales with length as $T \propto \sqrt{L}$. Differences from these predictions can arise from two main sources. First, the nonlinearity



Figure 1: Homemade pendulum setup consisting of 2 rulers, a chopstick, paper cup and tape

in Eq. (1) (the $\sin \theta$ term) produces amplitude-dependent periods at larger angles, so T may increase systematically with θ_0 . Second, the damping may not be well described by a single linear term $-\gamma\dot{\theta}$. The model also assumes that the apparatus is approximately symmetric, so releases to the left and to the right should yield consistent parameters.

In this experiment, a pendulum with adjustable effective length and release amplitude was constructed and recorded so that the angular displacement $\theta(t)$ could be extracted as a function of time. Using these data, I first tested for period dependence on amplitude by comparing T across release angles, second tested left-right symmetry by comparing results from opposite release directions, third, checked whether the decay envelope followed a single exponential and extracted τ , and finally, examined the scaling of T with L using a ratio comparison to the $T \propto \sqrt{L}$ prediction. An important outcome is therefore identifying the parameter range over which the damped, small-angle model is adequate within experimental uncertainty and the range over which systematic effects in the pendulum become significant.

2 Methods and Procedure

I built a pendulum 1 to test the damped harmonic oscillator model. The pivot consisted of two rulers and a chopstick: one ruler was rigidly taped to a tabletop with a small overhang past the edge, and a chopstick was taped on top of this ruler to serve as a cylindrical pin support. A second ruler (the moving arm) was suspended by placing one of its circular holes over the chopstick. The chopstick diameter was chosen to be comparable to the hole diameter so that the motion was mechanically guided and remained close to a single vertical plane. Because the hole pin tolerance still allowed small side to side play, a tensioned string was tied to both sides of the moving ruler such that it passed tightly at the hole location; this further prevented out of plane motion and reduced the tendency toward an elliptical

*julian.woo@mail.utoronto.ca

trajectory.

The bob was a paper cup which mass was measured on a scale prior to data collection. To vary the effective pendulum length while keeping the same bob, a ruler was taped inside the cup and used as a rigid reference for the string attachment point: attachment at the bottom of the cup defined the short configuration, attachment halfway up the cup defined the mid configuration, and attachment at the lip of the cup defined the long configuration. Data were collected for three lengths ($L \in \{30.2, 34.5, 37.1\}$ cm), three nominal release amplitudes, and three masses; however, the final analysis reported here uses a single mass in order to isolate the dependence on length and amplitude. For each configuration, multiple trials were recorded and symmetry was tested by releasing the pendulum from both sides: three trials released from the right and three from the left, while keeping the apparatus geometry and attachment point unchanged between left/right trials.

Nominal release angles were measured using a protractor and were defined operationally by specific contact/mark alignments of the ruler-cup assembly with the tabletop. The 138° condition corresponded to the configuration in which the 8 cm and 22 cm marks on the moving ruler were simultaneously in contact with the table (i.e., the ruler touches at those points at release). The 117° condition corresponded to the 10 cm and 20 cm marks touching the table at release. The 283° condition corresponded to the larger-angle release at which the cup itself first touched the table.

Motion data were acquired using the Physics Tracker app from video recordings. The camera was set to 1080p resolution at 60 fps and position was kept fixed for a given data-taking session and oriented so that the optical axis was approximately perpendicular to the plane of motion to reduce parallax. Tracker was used to follow the bob position frame-by-frame and export time-stamped coordinates (t, x, y) . The angular displacement time series $\theta(t)$ used in later analysis was obtained from these coordinates by treating the trajectory as an arc about a fixed pivot: a circle was fitted to the tracked (x, y) points to estimate the pivot/center (x_0, y_0) , the coordinates were shifted to $(x_c, y_c) = (x - x_0, y - y_0)$, and the angle was computed using a continuous phase convention (to avoid $\pm\pi$ discontinuities) before recentering the oscillation about zero. Periods were determined from the timing of successive angular maxima, and damping was characterized by the decay of the peak amplitude envelope over time.

2.1 Uncertainty and Control Variables

Several design and procedural choices were made specifically to control variables and reduce both random and systematic uncertainties in the extracted period and damping parameters. First, the pivot and suspension were built to restrict the motion to (approximately) two dimensions, since out-of-plane motion would project incorrectly into the tracked (x, y) coordinates and bias the derived $\theta(t)$. The moving ruler was hung on a chopstick “pin” whose diameter was chosen to be comparable to the ruler-hole diameter, mechanically guiding the motion into a single plane. Because the fit between the hole and pin was not perfectly tight, an additional taut string was tied to both sides of the moving ruler and tensioned so that it seated at the hole location; this further discouraged forward/backward wobble and reduced the tendency toward elliptical movement.

To isolate the effect of length and release amplitude, the bob geometry was held fixed while the effective length was varied. The bob was a paper cup, and a ruler was taped inside the cup to act as a rigid reference for defining the attachment location. Three repeatable attachment points were used (bottom of the cup for the shortest length, halfway

up for the mid length, and the cup lip for the longest length). Using a ruler reference reduces uncertainty in the effective length compared with a string as the attachment location can be reproduced to the millimetre scale using markings, and it is less sensitive to knot slip, slight changes in where the string seats on the bob, and string stretch. In contrast, tying directly to the cup without a rigid reference introduces ambiguity in the reference point and increases variation in the distance from pivot to the center of mass L , which would possibly weaken or bias any test of the prediction $T \propto \sqrt{L}$. Also a fully rigid arm such as a stiff rod could further reduce length variability).

Initial conditions were controlled to make amplitude dependence interpretable. For runs intended to compare T across amplitudes, the length configuration and bob mass were held fixed and only the release angle was varied. Release angles were set using a repeatable geometric rule (aligning the bob with specific marks before release) and the bob was released from rest to avoid adding an uncontrolled initial tangential velocity. For symmetry checks, left and rightside releases were performed under the same geometry: the attachment point, camera placement, and pivot configuration were kept unchanged, and trials were repeated (three releases from each side) so that any observed left-right difference could be attributed to apparatus asymmetry (for example pivot friction or residual 3-D motion) rather than a change in configuration.

Video recording choices were also made to reduce measurement uncertainty. The camera position was kept fixed during data collection and positioned as close as possible to perpendicular to the plane of motion to reduce perspective distortion in the tracked coordinates. Data was recorded at 60 fps, which decreases timing quantization in measurements. Finally, uncertainties on period points were constructed from two controlled contributions: the cycle to cycle scatter¹⁴ within a run and the frame-time resolution¹⁵ set by the video sampling interval. Combining these provides the error appropriate for weighted fits and for seeing whether deviations from the model exceed what is expected from measurement uncertainty.

3 Analysis and Discussion

3.1 Period versus amplitude

This portion of the experiment tested whether the pendulum’s oscillation period depended on amplitude. Video tracking provided the bob position $(x(t), y(t))$, which was converted into an angular coordinate $\theta(t)$ and analyzed to extract a period estimate T for each run.

The dataset consists of $N_{\text{pts}} = 36$ period measurements spanning amplitudes from approximately $\theta \approx 0.15$ to 2.0 rad (Fig. 2), grouped into three release angles (117, 138, 282). For each run, the angle time series was formed by fitting a circle to the tracked (x, y) points to determine the pivot/centre (x_0, y_0) , shifting to $(x_c, y_c) = (x - x_0, y - y_0)$, and computing $\theta(t) = [\arctan 2(x_c, -y_c)] - \langle \theta \rangle$. The period was extracted by locating successive positive peaks of $\theta(t)$ and forming periods by cycle $T_i = t_{i+1} - t_i$, with a mean period of $T_{\text{mean}} = \langle T_i \rangle$.

To test for amplitude dependence, we compared the measurements to the null model (constant period)

$$T(\theta) = T_0. \quad (4)$$

If the period depends on amplitude, we expect a systematic trend of T with θ , which would appear as structured (non-random) residuals as θ increases. The constant model was fit using a weighted mean (weights $w_i = 1/\sigma_{T_i}^2$), and the adequacy of the model was assessed using both residual plots and a χ^2 goodness-of-fit test.

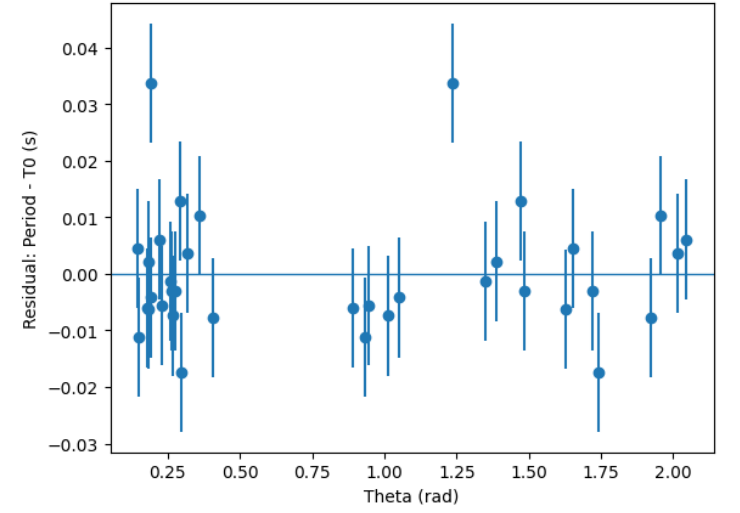
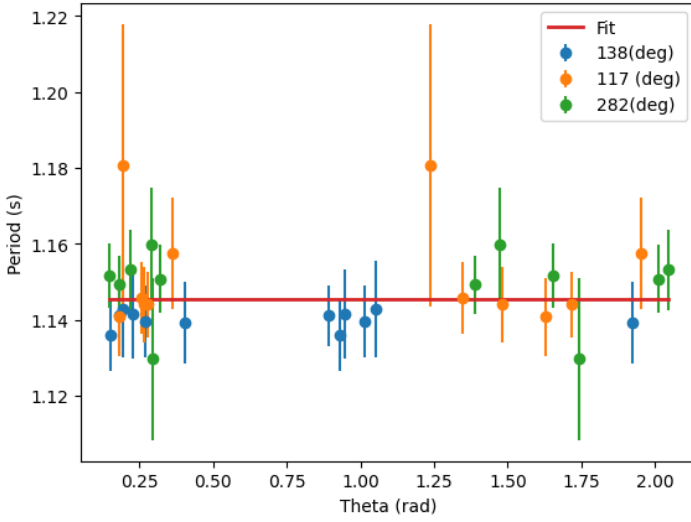


Figure 2: Period as a function of oscillation amplitude for three release-angle conditions, with vertical error bars showing the estimated uncertainty σ_T for each run. A weighted constant fit gives $T_0 = 1.15$ s, with goodness-of-fit $\chi^2 = 39.9$ and reduced $\chi^2_v = 1.14$ for $\nu = 35$ degrees of freedom.

Figure 3: Residuals $r_i = T_i - T_0$ plotted versus θ with the same σ_T error bars. The residuals are scattered about zero without a clear monotonic trend across the tested amplitude range, and their distribution is approximately symmetric about $r = 0$ (weighted mean residual $\bar{r}_w = (-1.7 \times 10^{-16} \pm 1.7 \times 10^{-3})$ s), indicating no detectable overall bias relative to the constant model. Together, the absence of a systematic residual trend with θ and $\chi^2_v \approx 1$ show that the data are consistent with no measurable amplitude dependence of the period at this sensitivity.

Uncertainties on each run's mean period were estimated from two contributions. First, the observed scatter within a run [14](#). Second, a timing-resolution contribution set by the Tracker time step [15](#) dt . These were combined [16](#), and used as the errors in [Fig 2](#) and [Fig. 3](#).

The best-fit constant period is

$$T_0 = 1.15 \text{ s.} \quad (5)$$

With $\nu = N_{\text{pts}} - 1 = 35$ degrees of freedom, the goodness-of-fit values are

$$\chi^2 = 39.9, \quad \chi^2_v = 1.14. \quad (6)$$

A reduced χ^2 modestly above one indicates that the constant model is broadly consistent with the data, with scatter only slightly larger than expected from the assigned uncertainties (uncertainties could also be mildly underestimated).

[Figure 3](#) shows the residuals $r_i = T_i - T_0$ plotted versus θ . The residuals are scattered about zero with no clear monotonic dependence across the full amplitude range, and the uncertainty bars frequently overlap $r = 0$. In addition, the residuals appear approximately symmetric about zero: there is no persistent tendency for points to lie predominantly above or predominantly below the constant-period model. This is supported by the weighted mean residual,

$$\bar{r}_w = (-1.7 \times 10^{-16} \pm 1.7 \times 10^{-3}) \text{ s,} \quad (7)$$

which is statistically indistinguishable from zero ($|z| \ll 1$). A small $\bar{r}_w$ indicates that positive and negative residuals balance on average, so the fit is not systematically biased high or low and is therefore consistent with symmetry about $r = 0$. These two diagnostics show different features of the data. Symmetry about zero (equivalently, $\bar{r}_w \approx 0$) tests for an overall offset in the fit whether the model tends to overestimate or underestimate T on average. By contrast, the absence of a trend with θ tests whether the period varies systematically with amplitude. A dataset can satisfy $\bar{r}_w = 0$ yet still show amplitude dependence if positive and negative residuals cancel when averaged.

Altogether, residuals consistent with zero within uncertainties, approximate symmetry about zero, and χ^2_v close to one these results support the notion that the constant-period model is adequate over the tested amplitude range.

3.2 Exponential decay test and decay constant at $L = 38.1$ cm

For a single pendulum length $L = 38.1$ cm, I tested verified that the oscillation amplitude decays exponentially in time and extracted the decay time constant τ .

From the tracked angle time series $\theta(t)$, we identified a sequence of envelope peak amplitudes

$$A_i = |\theta(t_i)|,$$

where t_i are the times of local maxima of $|\theta(t)|$ found via a python peak-finding routine with minimum peak separation set by an estimated period T . I repeated this procedure for three release angles (282° , 117° , 138°) at the same length, producing three datasets of (t_i, A_i) .

For a lightly damped oscillator, the angular displacement can be modeled as $\theta(t) = \theta_0 e^{-t/\tau} \cos(\frac{2\pi t}{T} + \phi)$, so the amplitude envelope should follow

$$A(t) = A_0 e^{-t/\tau}. \quad (8)$$

Taking the natural logarithm linearizes the model:

$$\ln A(t) = \ln A_0 - \frac{t}{\tau}. \quad (9)$$

Thus, an exponential decay is supported if $\ln(A_i)$ vs. t_i is consistent with a straight line and the log-space residuals show no systematic trend.

I fit the log-linear model

$$y_i \equiv \ln(A_i) = mt_i + c$$

over a time window where the signal to noise ratio remained adequate. The decay constant follows from the fitted slope: $\tau = -\frac{1}{m}$, $A_0 = e^c$.

Uncertainty in $\theta(t)$ arises from tracking/measurement [14](#) noise in the extracted $(x(t), y(t))$ coordinates and/or frame-to-frame [15](#) jitter in the estimated angle; we estimated an angle noise floor σ_θ from the high-frequency scatter of $\theta(t)$ about a smoothed version of the signal. Each envelope peak

amplitude was assigned $\sigma_{A,i} \approx \sigma_\theta$ (in radians), and this was propagated to log space using $\sigma_{y,i} = \sigma_{\ln A,i} \approx \frac{\sigma_{A,i}}{A_i}$. This naturally produces larger $\ln(A)$ error bars at late times when A_i becomes small. We performed a weighted linear fit in log space using weights $w_i = 1/\sigma_{y,i}^2$. The uncertainty in τ was obtained from propagation of the fitted slope uncertainty: $\sigma_\tau = \frac{\sigma_m}{m^2}$. As a goodness-of-fit check we computed the reduced chi-squared in log space:

$$\chi_v^2 = \frac{1}{N-2} \sum_{i=1}^N \left(\frac{r_i}{\sigma_{y,i}} \right)^2.$$

For $L = 38.1$ cm we obtained:

$$\begin{aligned} 282^\circ \text{ release: } \tau &= 15.4 \pm 0.4 \text{ s}, & \chi_v^2 &= 0.62, \\ 117^\circ \text{ release: } \tau &= 14.8 \pm 0.5 \text{ s}, & \chi_v^2 &= 0.79, \\ 138^\circ \text{ release: } \tau &= 14.8 \pm 0.5 \text{ s}, & \chi_v^2 &= 0.32. \end{aligned}$$

In all three runs, the $\ln(A)$ vs. t plots are approximately linear over the fitted region, and the log space residuals fluctuate about zero with most points consistent within their propagated uncertainties. Quantitatively, the reduced chi-squared values lies in the range $0.32 \leq \chi_v^2 \leq 0.79$, indicating that a exponential provides an adequate description of the decay over the analyzed interval. Therefore, for $L = 38.1$ cm the amplitude decay is consistent with exponential behavior within the quoted uncertainties, and a representative value is $\tau \approx 15$ s. All fits yield $\chi_v^2 < 1$, suggesting the assigned uncertainties are somewhat conservative. In either case, these values support the conclusion that the exponential model is not being forced onto incompatible data. Potential systematic effects include: (i) reduced accuracy of peak identification at late times when A_i approaches the noise floor; (ii) minor out of plane motion affecting the tracked angle; (iii) slight changes in effective damping due to pivot friction effects. These would most likely appear as structured residuals at late times; however, given the increasing $\sigma_{\ln A}$ at small A_i , no statistically significant breakdown of the exponential model is found.

3.3 Does the period depend on length?

In this part, I verified the model prediction that the oscillation period scales with the square root of the effective length, $T = C\sqrt{L_{\text{eff}}}$, $L_{\text{eff}} = L + D$,

I used three lengths and collected six repeated datasets per configuration. For each dataset, the tracked $(x(t), y(t))$ coordinates were converted into an angular time series $\theta(t)$ by first fitting a circle to determine the pivot/rotation center and then computing $\theta(t) = (\arctan 2(x - x_0, -(y - y_0)))$. The period for a single dataset was estimated from the times of successive positive peaks of $\theta(t)$: if t_k are peak times, the cycle periods are $T_k = t_{k+1} - t_k$, and I used

$$T_{\text{mean}} = \langle T_k \rangle, \quad T_{\text{std}} = \text{std}(T_k). \quad (10)$$

Thus, each of the three configurations produced a set of six measured mean periods $\{T_{\text{mean},j}\}$ with an uncertainty estimate $\sigma_{T,j}$.

For each dataset, I estimated the period uncertainty by combining scatter in the cycle-to-cycle 14 periods within that dataset and a timing-resolution¹⁵ term tied to the sampling interval Δt . With N measured cycles in that dataset,

I then computed a weighted mean period $\bar{T}$ for each length using weights $w_j = 1/\sigma_{T,j}^2$ and uncertainty $\sigma_{\bar{T}} = (\sum_j w_j)^{-1/2}$. This approach treats repeated trials as independent and assumes the dominant uncertainties are captured by scatter and timing resolution.

I used two complementary, quantitative tests of the square-root scaling

$$T = C\sqrt{L_{\text{eff}}}:$$

1. **Pairwise ratio test** For any two lengths L_i, L_j ,

$$\frac{T_i}{T_j} = \sqrt{\frac{L_i}{L_j}} \implies R_{ij} \equiv \frac{(T_i/T_j)}{\sqrt{L_i/L_j}} = 1. \quad (11)$$

This cancels the unknown constant C , so it tests the functional dependence directly.

2. **Linearized fit.** A plot of T^2 vs. L_{eff} should be a straight line with slope $a = C^2$ and intercept $b = 0$ if L_{eff} has been defined correctly. I therefore fit

$$T^2 = aL_{\text{eff}} + b \quad (12)$$

with weighting by $\sigma_{T^2} = 2T\sigma_T$, and I inspected the residuals and χ_{red}^2 as required.

The ratio test provides a dimensionless check of the $\sqrt{L}$. The linearized T^2 vs L_{eff} fit provides a second check and also tells me whether the effective-length definition is consistent through the intercept b . Using the weighted mean period at each length, I computed the pairwise ratio statistic R_{ij} . The results were:

- Longest vs Shortest: $R = 0.954 \pm 0.006$, $z = (R - 1)/\sigma_R = -7.78$
- Longest vs halfL: $R = 0.980 \pm 0.006$, $z = -3.46$
- Shortest vs halfL: $R = 1.03 \pm 0.01$, $z = +3.99$

All three comparisons deviate from the prediction $R = 1$ by more than 3σ , and one deviates by nearly 8σ . Therefore, with the uncertainties as currently estimated, the dataset refutes the claim $T \propto \sqrt{L_{\text{eff}}}$. This is a quantitative refutation because it is based on multiple independent pairwise tests.

The weighted fit to $T^2 = aL_{\text{eff}} + b$ returned a nonzero intercept ($b \neq 0$ at many standard deviations in my output), which is inconsistent with ideal expectations $b = 0$. In practice, a large b indicates that at least one of the following is true: L_{eff} is missing an offset, is not properly included or has the wrong sign/magnitude, or the length measurements do not correspond to the true pivot-to-center-of-mass distance, or there is a systematic problems in the extracted periods that does not scale as $\sqrt{L}$. The residual plot for this fit is the appropriate method for determining whether deviations are random scatter or structure.

Taken at face value, the ratio test already refutes the $\sqrt{L_{\text{eff}}}$. However, the direction and pattern of deviations (we see some ratios below 1 and one above 1) suggests that the dominant issue may be systematic error rather than random noise. The most likely contributors are:

Firstly, **Effective length definition (D and center-of-mass).** The model statement uses $L_{\text{eff}} = L + D$, where D is the distance from the string attachment point on the bob to its center of mass. If D is not measured and included, then the mapping from configuration to L_{eff} can be wrong by a length-independent offset, which produces an apparent nonzero intercept in T^2 vs L and can distort the pairwise ratios.

Secondly, **Amplitude control.** The project emphasizes checking amplitude dependence separately. If the initial amplitudes differ between lengths, then even if the true physics preserves a $\sqrt{L}$ scaling at fixed amplitude, the measured ratios can deviate because the multiplicative amplitude factor is not constant across configurations. This is especially important when operating at large angles, which the project explicitly encourages for amplitude tests.

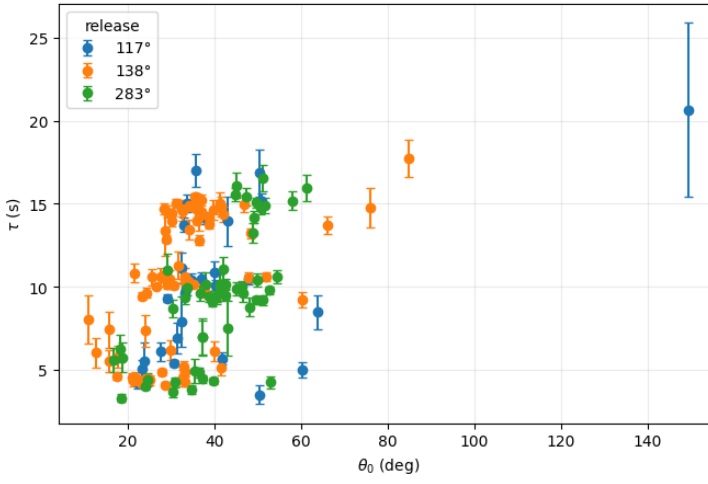


Figure 7: τ vs. measured initial amplitude θ_0 . Each point shows the decay time constant τ extracted from a single trial by fitting an exponential amplitude envelope, plotted against the corresponding initial amplitude θ_0 . Points are colored by release-angle category (117° , 138° , 282°). Vertical error bars represent the uncertainty on τ from the log-envelope regression for that trial. The scatter forms clusters primarily associated with release-angle group and configuration rather than a single monotonic $\tau(\theta_0)$ relation across the full dataset.

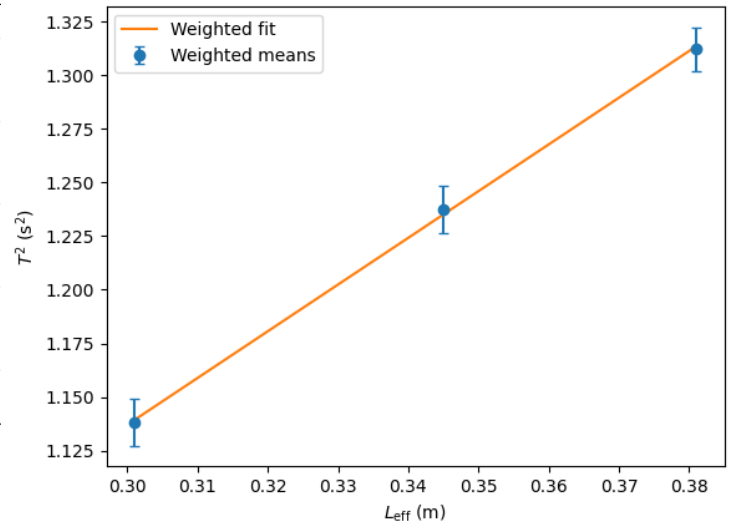


Figure 8: Weighted least-squares fit of the linearized model $T^2 = aL_{\text{eff}} + b$ to the three weighted-mean points of Fig. 14, using weights $1/\sigma_{T^2}^2$. The best-fit parameters are $a = 2.18 \pm 0.05 \text{ s}^2/\text{m}$ and $b = 0.484 \pm 0.018 \text{ s}^2$ (uncertainties 1σ); the goodness-of-fit is $\chi_{\text{red}}^2 = 0.08$ for $\text{dof} = 1$. The nonzero intercept b indicates that this dataset does not follow the idealized zero-intercept form over the measured range within the stated uncertainties (suggesting an unaccounted effective-length offset and/or systematic effects).

Finally, **Underestimated uncertainty** The current σ_T model includes within-run scatter and timestamp resolution, but it may not include trial-to-trial effects such as: slight out-of-plane motion, changing pivot friction, changes in release technique, or biases from peak-picking when the waveform is not perfectly sinusoidal. The writing guidelines emphasize that uncertainty discussion must include both measurement and experimental uncertainty; here the experimental/systematic terms likely dominate.

The proof against is quantitatively strong because the deviations are several standard deviations away from one. The next steps should be: (i) measure and include D so L_{eff} is truly pivot-to-center-of-mass, (ii) ensure the same initial amplitude (or restrict to a narrow amplitude band) for all lengths when testing length scaling, and (iii) add a systematic uncertainty floor (or include length uncertainties) so the error bars reflect trial-to-trial variability.

Using the three lengths and a ratio-based test, the measurements give $R_{ij} \neq 1$ at 3σ to 8σ across all length pairs. Therefore, with the present L_{eff} definition and uncertainty model, the claim $T = C\sqrt{L_{\text{eff}}}$ is refuted by the data.

The most likelt explanation is unaccounted systematic error, and improving those controls is required before attributing the discrepancy to a failure of the physical model.

Empirical dependence of τ on $(L+D)$ and θ_0 The purpose of this analysis was to search for empirical trends and evaluate simple fit functions using goodness-of-fit statistics. In total, 159 trial-level τ values were extracted across three $(L+D)$ settings (0.301 m, 0.345 m, 0.381 m) and three release-angle categories (117° , 138° , 283°), with repeated trials at each configuration.

Throughout, τ is defined as the $1/e$ decay time of the amplitude envelope: $A(t) = A_0 e^{-t/\tau}$ (equivalently $\ln A = -t/\tau + \ln A_0$). For each trial, the angular time series $\theta(t)$ was constructed from the tracked motion and an amplitude envelope was obtained by sampling peak values of $|\theta|$ over time. An exponential decay was fit in log space,

$$\ln |\theta| = mt + c, \quad \tau = -\frac{1}{m}, \quad (13)$$

using weighted linear regression in $\ln |\theta|$ with uncertainties propagated into log space. A representative extraction is

shown in Fig. 9. In that example, the data is well-described by a straight line in $\ln |\theta|$ with residuals that show no strong systematic structure, and the fit quality is decent ($\chi_{\text{red}}^2 = 0.691$, $p = 0.909$, $\text{dof} = 33$). This diagnostic is important because it shows that an exponential envelope provides a reasonable description for an individual run over the chosen window.

To compare configurations, repeated trials were summarized using weighted means $\bar{\tau}$ for each {release angle, $(L+D)$ } group. Figure 12 plots $\bar{\tau}$ versus $(L+D)$ with error bars. The three release-angle categories form clearly separated bands at fixed $(L+D)$, indicating that τ is not determined by length alone in this dataset. In addition, within each release-angle category the mean values change across the three $(L+D)$ settings (e.g., for 138° the mean decreases from $\bar{\tau} \approx 10.8 \text{ s}$ at $(L+D) = 0.301 \text{ m}$ to $\bar{\tau} \approx 8.36 \text{ s}$ at 0.345 m , and remains near $\bar{\tau} \approx 8.11 \text{ s}$ at 0.381 m), showing qualitative configuration dependence rather than a single length-independent value.

Only basic functions were tested: constant fits ($\tau = a$) and linear fits ($\tau = a + b(L+D)$), performed separately within each release angle category. Goodness-of-fit was assessed using χ^2 , χ_{red}^2 , and the associated p -value, and by inspecting residual plots. Using the fit-derived uncertainties on τ as the point-by-point σ_τ , constant fits to the weighted means were strongly rejected for every release-angle: 117° : $\chi_{\text{red}}^2 = 278$ ($\text{dof} = 1$, $p \approx 0$); 138° : $\chi_{\text{red}}^2 = 769$ ($\text{dof} = 2$, $p \approx 0$); 282° : $\chi_{\text{red}}^2 = 141$ ($\text{dof} = 2$, $p \approx 0$). Linear fits likewise did not resolve the discrepancy for the two subsets with three distinct $(L+D)$ values: 138° yields $\chi_{\text{red}}^2 = 103$ ($\text{dof} = 1$, $p \approx 0$), and 282° yields $\chi_{\text{red}}^2 = 58.9$ ($\text{dof} = 1$, $p = 1.67 \times 10^{-14}$). For the 117° subset there are only two distinct $(L+D)$ values, so a two-parameter linear fit has $\text{dof} = 0$ and χ_{red}^2 (and p) are not defined; this subset can show a slope but cannot be used to assess goodness-of-fit or discriminate functional form. Representative fit residuals are shown in Fig. 13, where the deviations are large compared with the quoted error bars.

The combination of acceptable exponential-envelope fits within individual trials (Fig. 9) and extreme over-rejection

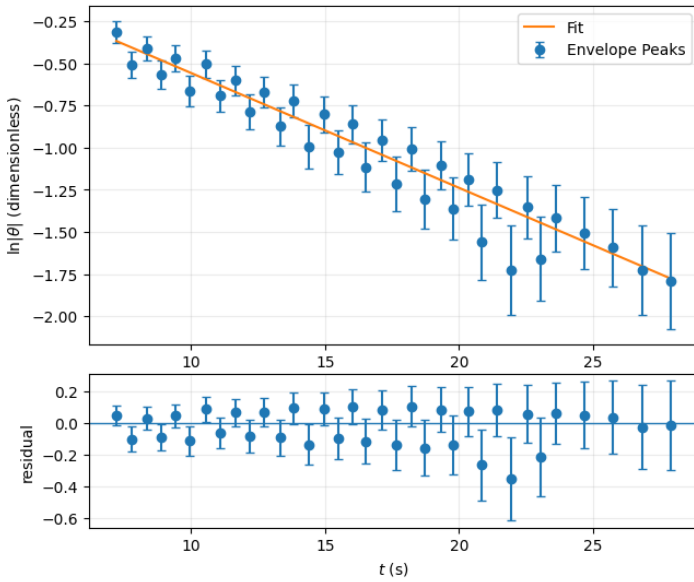


Figure 9: Extraction of τ from a log envelope fit. Envelope peak amplitudes $A(t) = |\theta(t)|$ were sampled at successive peaks and fit to an exponential decay in log space, $\ln A = mt + c$, giving $\tau = -1/m$ (top panel). Error bars show the propagated uncertainty in $\ln A$ for each envelope peak. The lower panel shows residuals $\ln A - (mt + c)$ with a zero reference line, used to assess systematic deviations from exponential decay. For this run, the residuals show no strong remaining structure and the fit is adequate

of constant/linear models for configuration means indicates that the trial-to-trial spread in τ is much larger than the uncertainty estimates produced by the envelope-fit regression alone. In other words, the regression-derived σ_τ reflects the within-run scatter of the envelope points, but it does not capture dominant run-to-run variability. Consequently, although Fig. 12 demonstrates qualitative dependence of τ on $(L+D)$ and clear separation by release angle grouping, the current dataset does not support a statistically acceptable low-parameter fit (constant or linear) under the regression-only uncertainty model.

Figure 7 plots trial-level τ versus the measured initial amplitude θ_0 , where θ_0 was defined as a early-time amplitude estimate (the median of the first few detected peak amplitudes). The points do not follow a single monotonic relationship across all trials; instead, they form clusters likely associated with release angle and $(L+D)$ configuration, with additional scatter and occasional outliers. Therefore, within the sensitivity of the present analysis, any dependence of τ on θ_0 cannot be isolated cleanly from configuration dependence and run to run variability.

Overall, τ exhibits configuration dependence in both $(L+D)$ and release-angle, but simple constant/linear models are rejected when uncertainties are taken solely from envelope-regression, implying that unmodelled run-to-run effects dominate the observed scatter in τ .

4 Conclusion

This project tested how well a damped simple-pendulum model describes a home-built pendulum by measuring the oscillation period T and decay time constant τ while varying the release amplitude and effective length, and by checking for left-right symmetry.

Over the tested amplitude range ($\theta \approx 0.15$ to 2.0 rad), the period measurements are consistent with a constant-period (no amplitude-dependence) null model. A weighted constant fit gives $T_0 = 1.15$ s, with a reduced chi-squared $\chi^2_\nu = 1.14$, indicating that the overall scatter is close to what is expected from the assigned uncertainties. The

residuals about T_0 show no systematic trend with θ and frequently overlap zero within error bars, so there is no statistically resolvable amplitude dependence in T for this dataset. In addition, the residuals are approximately symmetric about zero, supported by the weighted mean residual $\bar{r}_w = (-1.7 \times 10^{-16} \pm 1.7 \times 10^{-3})$ s (i.e., $|z| \ll 1$), which indicates that the constant-period fit is not biased high or low on average and is consistent with symmetry about $r = 0$.

The amplitude decay was also found to be consistent with a single-exponential envelope within uncertainty for the runs analyzed. Fitting the envelope to $A(t) = A_0 e^{-t/\tau}$ at fixed length yields representative values near $\tau \sim 15$ s; for example, at $L = 38.1$ cm the fitted time constants $\tau = 15.4 \pm 0.4$ s and $\tau = 14.8 \pm 0.5$ s (for two release-angle conditions) have reduced chi-squared values below one, which supports the adequacy of an exponential model given the adopted amplitude and log-space uncertainties. While late-time peak identification and residual out-of-plane motion can plausibly introduce systematic structure, no statistically decisive breakdown of the single-exponential model is observed within the sensitivity of the present analysis.

In contrast, the predicted square-root scaling of period with length is refuted by the length-dependence tests as implemented here. Pairwise ratio tests give values significantly different from unity, for example $R_{30.2,34.5} = 0.954 \pm 0.003$, $R_{30.2,37.1} = 0.927 \pm 0.003$, and $R_{34.5,37.1} = 0.972 \pm 0.003$, which are far outside the $\sim 2\sigma$ consistency band expected if $T \propto \sqrt{L}$. This disagreement most likely reflects unmodelled systematics in the effective-length definition or geometry-dependent factors (such as, small 3-D motion, pivot behaviour, or configuration dependent changes in the pivot-to-centre-of-mass distance). A design choice that reduced one important systematic was to define the bob attachment using a rigid ruler reference inside the cup, which improves repeatability of the attachment location compared with tying directly to the cup and helps control run-to-run variation in L . Even so, the length test remains sensitive to residual ambiguity in the true effective length (including any offset D), so improving the apparatus further (e.g., a more rigid pivot/hinge, tighter constraint to planar motion, and a direct geometric measurement of the pivot to centre-of-mass distance for each configuration) would be expected to strengthen the length scaling test and clarify whether the observed refutation is physical or from the experiment instrument.

Finally, no single simple dependence of τ on length and amplitude could be isolated cleanly: τ values cluster by configuration and release angle with additional scatter and outliers, and simple constant/linear empirical models perform poorly when uncertainties are taken only from the envelope regression method. Future work should therefore prioritize reducing configuration dependent systematics and adopting a more complete uncertainty model for τ that captures run-to-run effects. A practical identification of whether dissipation is small enough for clean exponential envelope measurements is the ratio τ/T ; if τ is not at least $\sim 20T$, friction or damping will be large and the apparatus should be redesigned because decay data becomes dominated by losses rather than ideal model behaviour. With these improvements, the same analysis framework (weighted fits, residual checks, and goodness-of-fit statistics) can better map the system and identify which specific experimental limitations drive departures from the ideal model.

Statement of AI Usage only autocomplete from Vscode and the table of equations in the introduction, no other AI usage

References

- [1] PHY324 Pendulum Project – 2025. Course handout, University of Toronto Department of Physics (PHY324), 2025.
- [2] Brown, D. *Tracker Video Analysis and Modeling Tool*. Open Source Physics (OSP).

Appendix

Uncertainty

Cycle-to-cycle scatter within a run:

$$\sigma_{T,\text{scatter}} = \frac{T_{\text{std}}}{\sqrt{N}}, \quad (14)$$

Timing-resolution contribution set by the Tracker time step dt ,

$$\sigma_{T,\text{frame}} \approx \frac{dt}{\sqrt{2N}}. \quad (15)$$

These were combined in quadrature,

$$\sigma_T = \sqrt{\sigma_{T,\text{scatter}}^2 + \sigma_{T,\text{frame}}^2} \quad (16)$$

Plots

Below and on the right

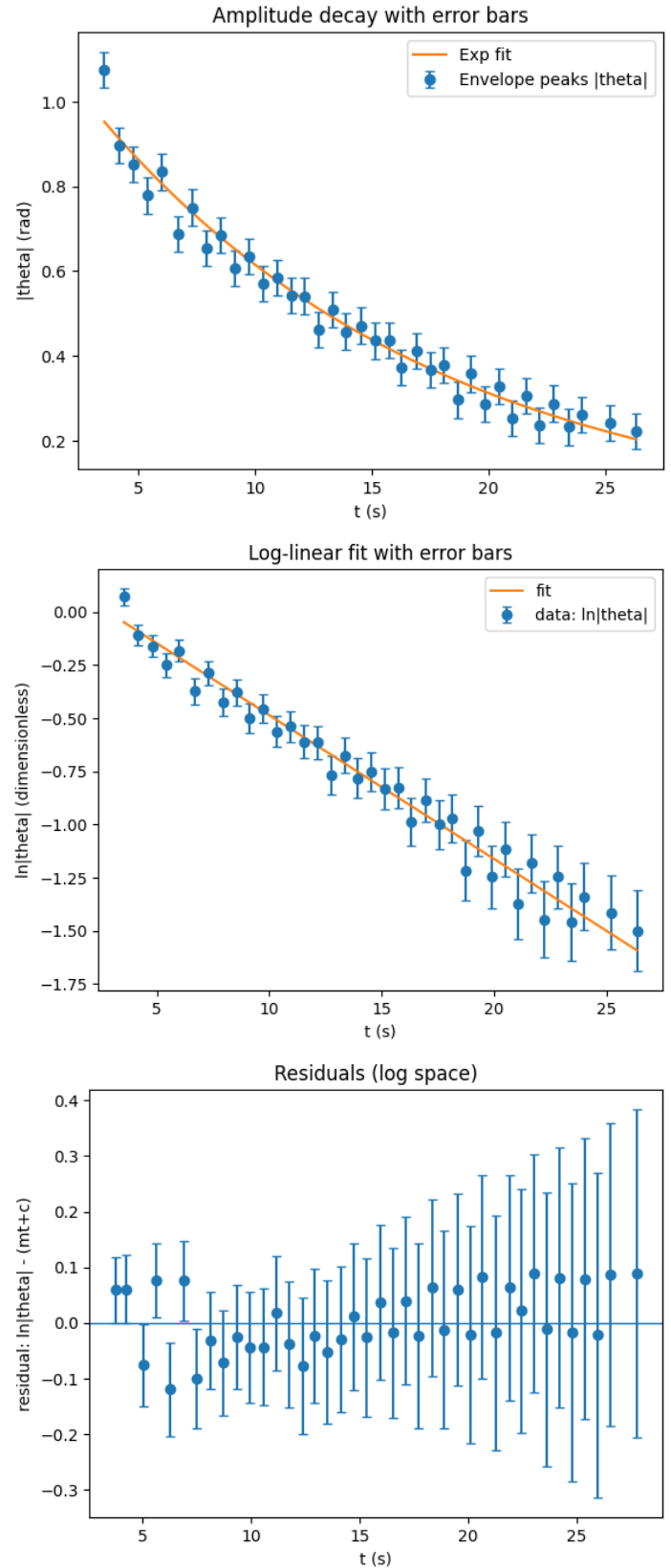


Figure 10: Amplitude decay at fixed length $L = 38.1$ cm (release angle 282°) **Top** Envelope peak amplitudes $A(t) = |\theta(t)|$ extracted from the tracked angle time series, with best-fit exponential model $A(t) = A_0 e^{-t/\tau}$ over the selected fit range. **Middle** Log-linear test of exponential decay: $\ln A$ vs. t with the weighted best-fit line $\ln A = mt + c$ (so $\tau = -1/m$). **Bottom** Log-space residuals $r_i = \ln(A_i) - (mt_i + c)$ used to assess model adequacy. Vertical error bars in (b,c) are propagated from the amplitude uncertainty via $\sigma_{\ln A} \approx \sigma_A / A$, producing larger uncertainties at late times when A is small. For this run, the fitted time constant is $\tau = 15.4 \pm 0.4$ s with reduced chi-squared $\chi^2_v = 0.62$ (dof = 38), indicating the data are consistent with a single-exponential envelope within uncertainties.

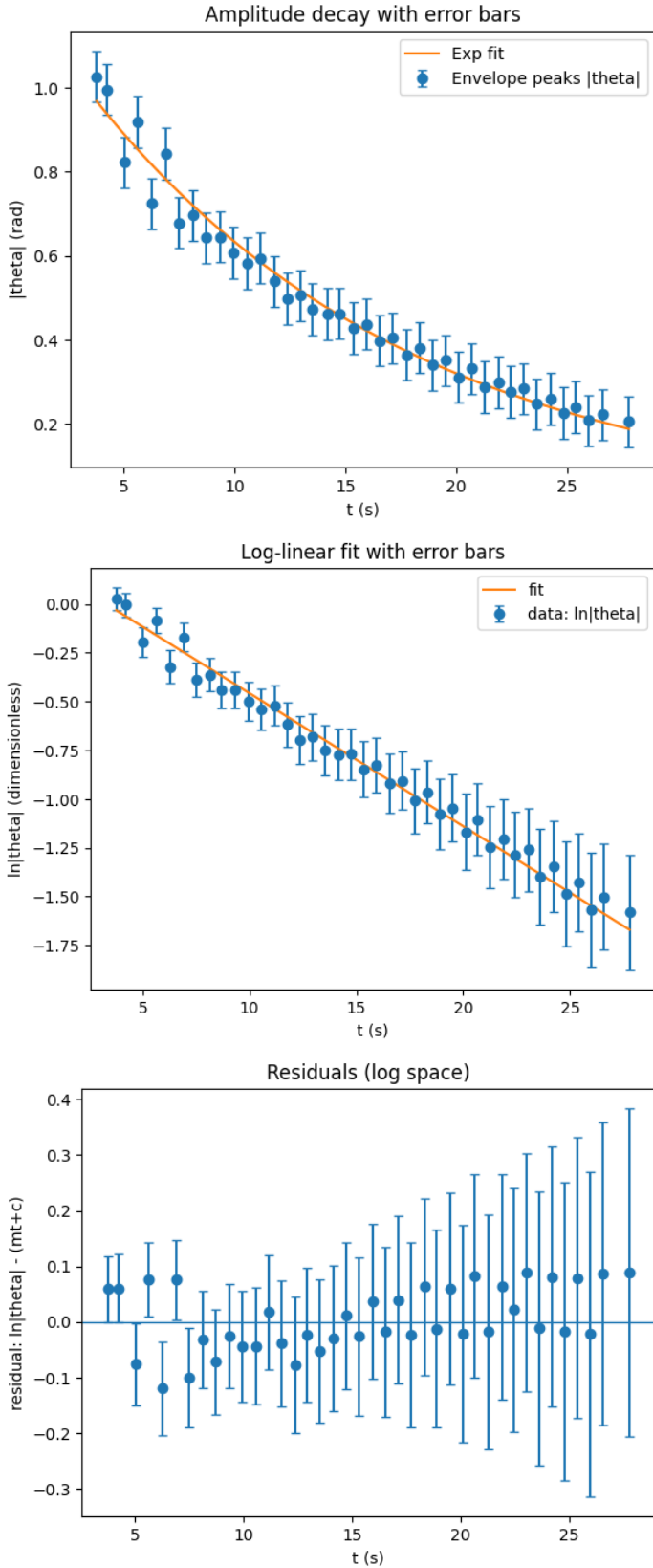


Figure 11: decay at fixed length $L = 38.1$ cm (release angle 117°). **Top** Envelope peak amplitudes $A(t) = |\theta(t)|$ with exponential fit $A(t) = A_0 e^{-t/\tau}$. **Middle** Log-linear representation $\ln A$ vs. t with the weighted best-fit line $\ln A = mt + c$. **Bottom** Log-space residuals $r_i = \ln(A_i) - (mt_i + c)$. Error bars are propagated into log space using $\sigma_{\ln A} \approx \sigma_A / A$, so late points carry larger fractional uncertainty and less weight in the fit. The fitted decay constant is $\tau = 14.8 \pm 0.5$ s with $\chi^2_v = 0.79$, supporting an exponential decay model for this dataset within the assigned uncertainties.

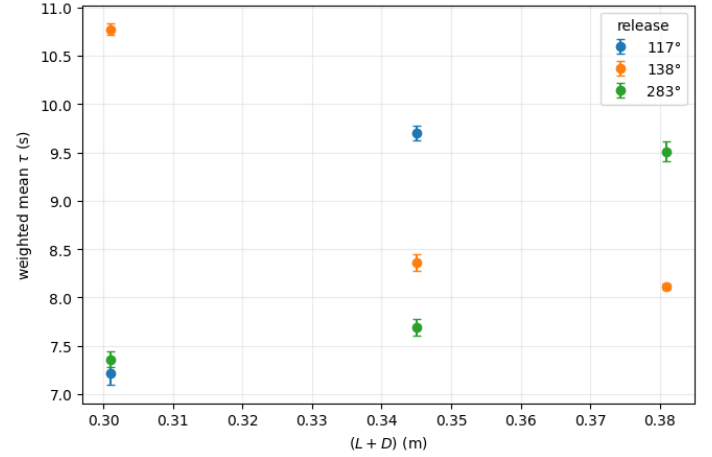


Figure 12: Weighted-mean $\bar{\tau}$ vs. effective length scale $(L+D)$. For each $\{\text{release angle}, (L+D)\}$ configuration, repeated trials were combined into a weighted mean $\bar{\tau}$ using inverse-variance weights from the individual trial uncertainties. Points are colored by release-angle category (117° , 138° , 283°). Vertical error bars indicate the standard error on the weighted mean. The separated bands at fixed $(L+D)$ show that τ depends on release-angle grouping in addition to $(L+D)$, and the variation across the three $(L+D)$ values indicates configuration dependence rather than a single length-independent τ .

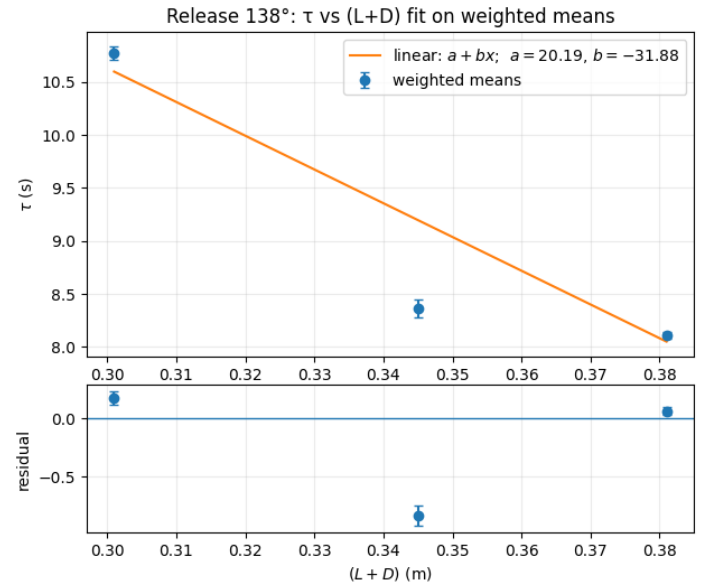


Figure 13: Example empirical model test: linear fit of $\bar{\tau}$ vs. $(L+D)$ for the 138° subset. Weighted-mean values $\bar{\tau}$ for the 138° release-angle category (points with error bars) were fit to a linear function $\tau = a + b(L+D)$ (top panel). The lower panel shows residuals about the fit with a zero reference line. Despite capturing the overall decrease between the smallest and largest $(L+D)$ values, the residuals remain large compared with the uncertainties, consistent with the poor goodness-of-fit statistics

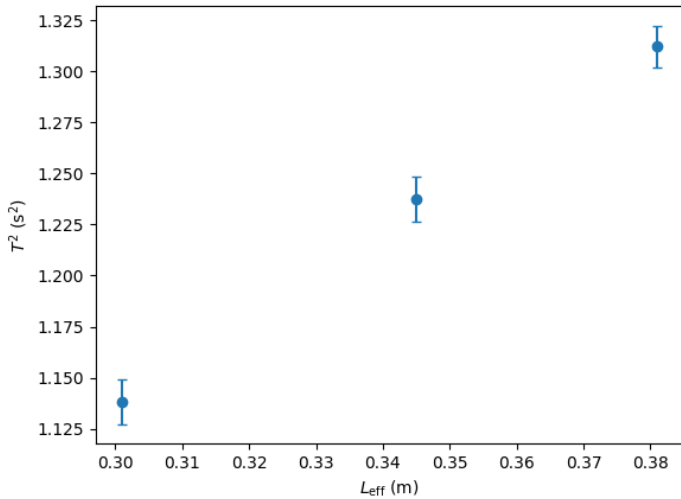


Figure 14: The same three configuration means as in Fig. 15, replotted as $T^2 \text{ (s}^2\text{)}$ versus $L_{\text{eff}} \text{ (m)}$. If $T \propto \sqrt{L_{\text{eff}}}$, then T^2 should be linear in L_{eff} . Error bars are 1σ uncertainties propagated from the period uncertainties via $\sigma_{T^2} = 2T \sigma_T$.

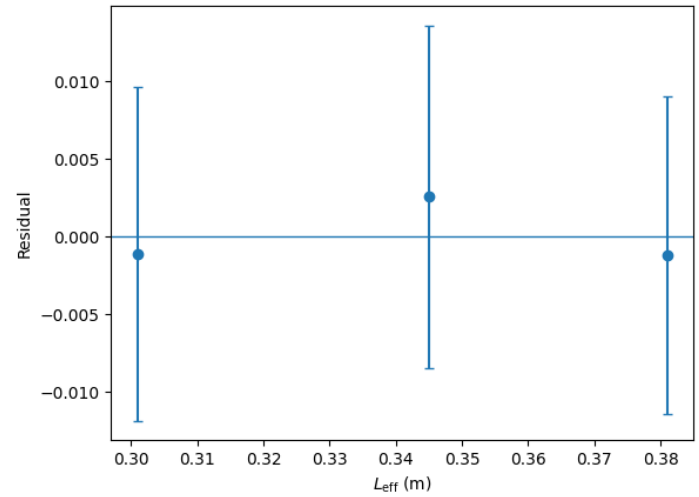


Figure 16: Residuals of the weighted linear fit in Fig. 8, defined as $\Delta = T_i^2 - (a L_{\text{eff},i} + b)$, plotted versus L_{eff} . Error bars show the 1σ uncertainties σ_{T^2} for each point. The residuals are consistent with zero within uncertainties for all three lengths; the very small χ^2_{red} suggests the scatter about the fit is smaller than expected from the uncertainties

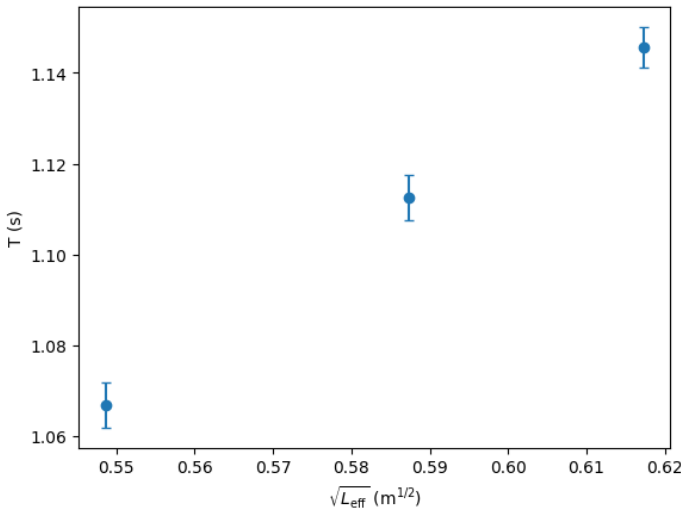


Figure 15: Measured pendulum period $T \text{ (s)}$ plotted against $\sqrt{L_{\text{eff}}} \text{ (m}^{1/2}\text{)}$ for three length configurations. Each point is the weighted mean period from six repeated tracker datasets for that configuration (one point per length), and the vertical error bars show the 1σ uncertainty on the weighted mean (computed from the per-dataset period uncertainties). The effective length is $L_{\text{eff}} = L + D$ (if $D = 0$ then $L_{\text{eff}} = L$). Plotting T vs $\sqrt{L_{\text{eff}}}$ linearizes the model prediction $T \propto \sqrt{L_{\text{eff}}}$; the increase of T with $\sqrt{L_{\text{eff}}}$ is consistent with that qualitative trend.